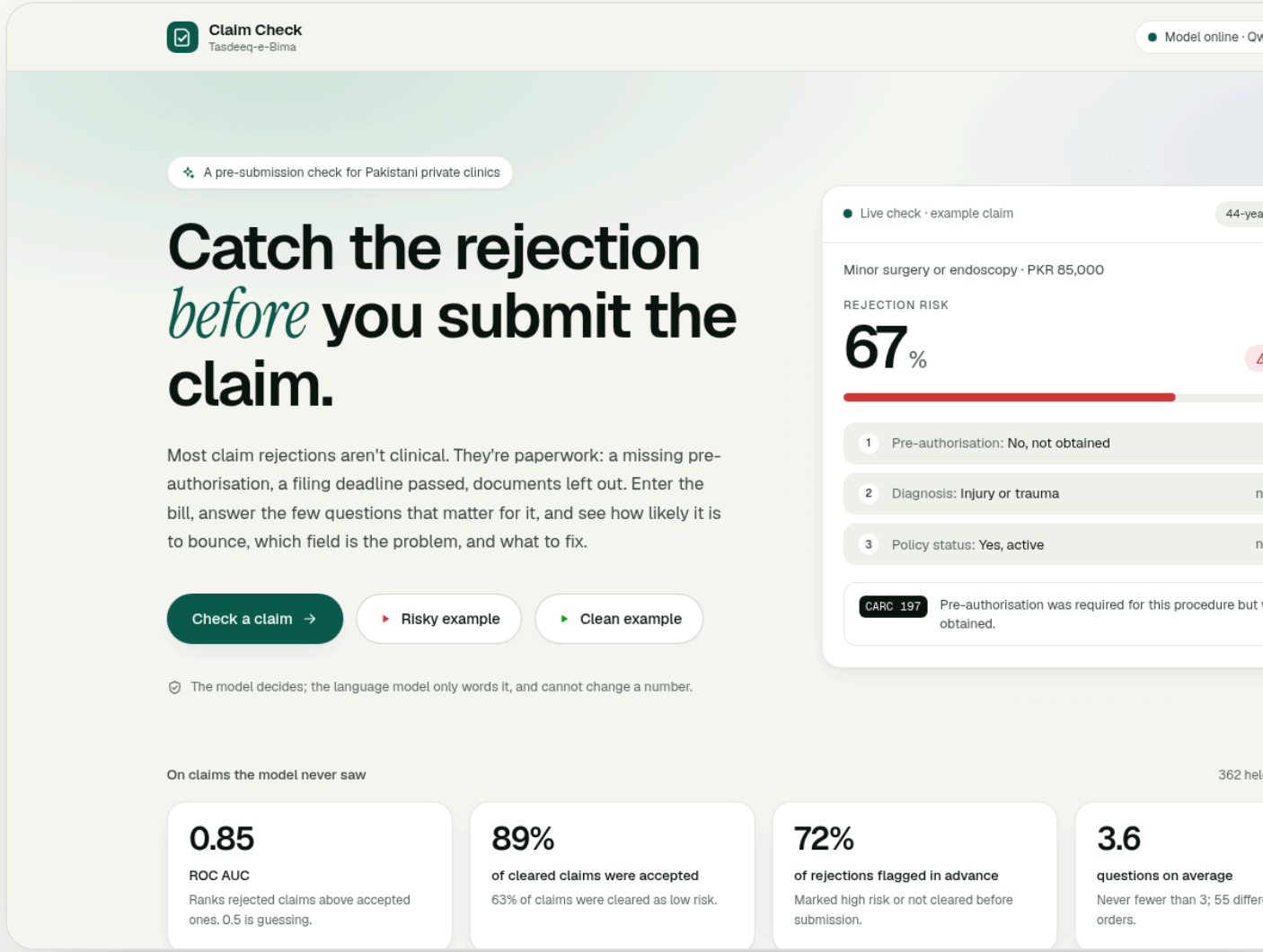
BANO QABIL AI HACKATHON

Claim Check

Catch the insurance-claim rejection *before* the claim is sent.

Mirza Ahsan Baig · Safee Akmal
Tasdeeq-e-Bima



THE PROBLEM

Claims get rejected for paperwork, not medicine

- A clinic treats an insured patient, then bills the insurer
- Weeks later the claim bounces: **no pre-authorisation**, a **missed filing deadline**, **missing documents**, an **amount over tariff**
- Staff rework and resubmit while the money is stuck
- Small clinics have no billing specialist to catch it

PRE-AUTHORISATION

Insurer approval needed **before** a planned treatment. Miss it and correct, necessary care can still go unpaid.

94% of US physicians say it delays care.[9]

Sources: [9] American Medical Association

GLOBAL SCALE · UNITED STATES

In the US alone, fighting denials costs $25.7 billion a year

15%

of claims to private insurers denied at first submission[1]

$25.7B

spent by US providers fighting denials in 2023[1]

70%

of those denials were paid in the end — after costly review[1]

86%

of denials are potentially avoidable[2]

On ACA marketplace plans, insurers denied about **1 in 5** in-network claims in 2023, and **fewer than 1%** of denials were appealed.[3]

Sources: [1] Premier Inc. · [2] Becker's Hospital Review · [3] KFF

GLOBAL SCALE · EMERGING MARKETS

Hardest where insurance is new

INDIA

71%

of health-claim value actually paid in FY 2023-24. **₹26,000 crore** disallowed or repudiated.[4]

PAKISTAN · KP SEHAT CARD

5–8 mo

payment waits for some hospitals. The programme blamed **incomplete documents**; 94% of hospitals asked for e-claims.[5]

KENYA · SHA

600+

rural private hospitals stopped accepting the new national cover; 89% called the claims portal non-functional.[6]

Sources: [4] Outlook Business · [5] Aaj News · [6] Business Daily Africa

WHY IT MATTERS

Avoidable rejections are a financial-inclusion problem

- Governments are scaling health insurance to close the coverage gap
- That means millions of new claims, new rules — and new rejections
- Clinics that aren’t paid stop accepting insured patients
- The patients insurance was meant to protect lose access

4.6B

people lack essential health services[7]

2.1B

face financial hardship paying for care[7]

Sources: [7] WHO & World Bank

THE OPPORTUNITY

Big hospitals have claim checkers. Small clinics don’t.

- Claim-checking software exists — built into large hospital billing systems
- A small clinic in a new insurance market has a receptionist and a form
- No tool tells them, before submitting, **what will bounce and why**

≈$15B

denial-management market, 2024[8]

≈$30B

projected by 2030 (industry estimate)[8]

Sources: [8] MarkNtel Advisors

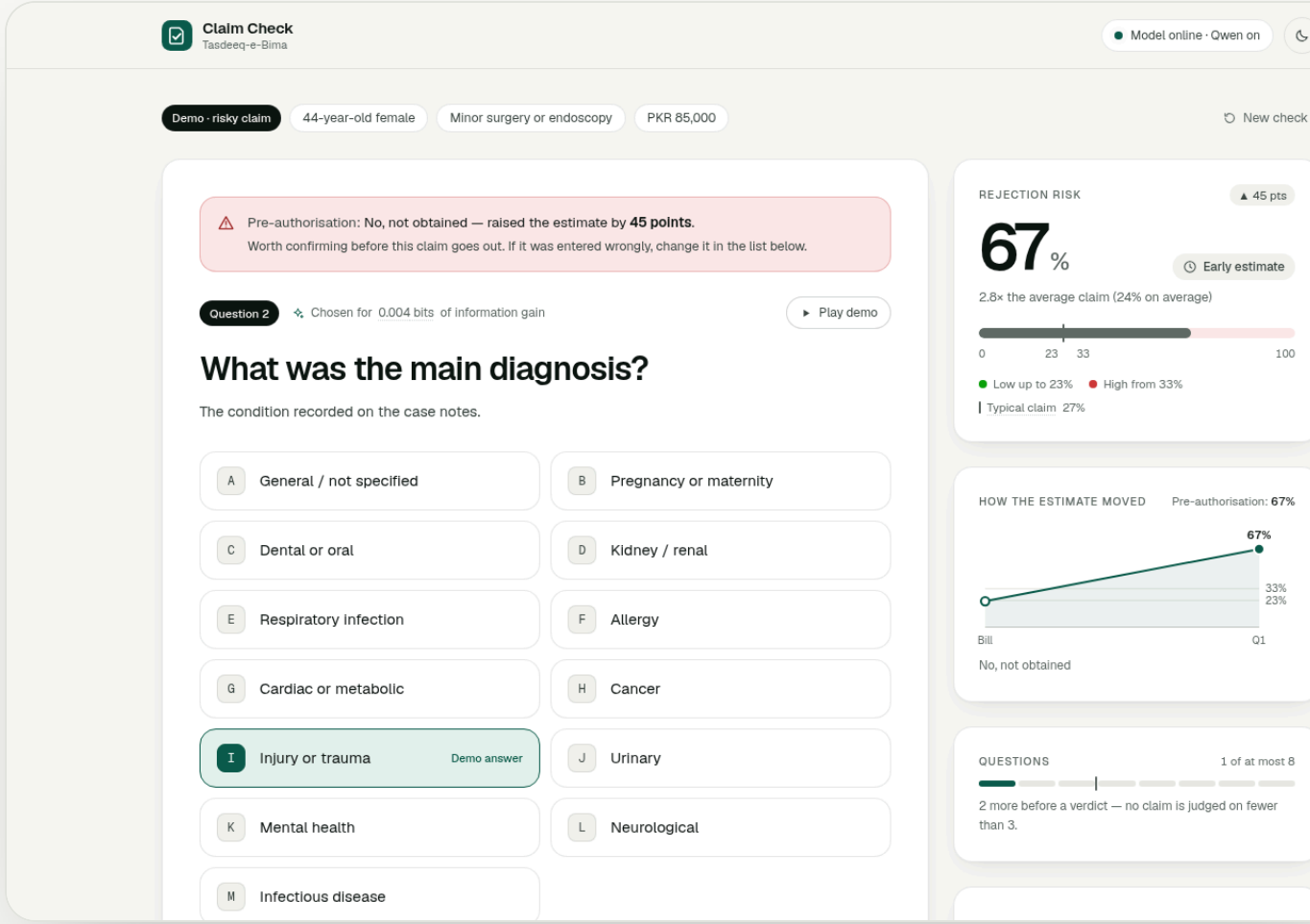
OUR SOLUTION

A spell-checker for insurance claims

1. **Enter the bill**
   Patient age and sex, procedure, amount

2. **Answer 3–4 questions**
   Only the ones that matter for this claim

3. **Fix before sending**
   Chance of rejection, the cause, the fix



BUSINESS CASE

Everyone in the claim wins

CLINICS

- Paid sooner
- Less rework
- Predictable cash flow

INSURERS & TPAS

- Cleaner claims
- Fewer appeals to process
- Lower admin cost

PATIENTS

- Clinics keep accepting their insurance
- Care isn’t held up by paperwork

How it could earn (proposed): per-clinic subscription · licence to TPAs for their provider panel · integration with national e-claims schemes

PART TWO

How it works

HOW IT WORKS

It asks only the questions that matter

- Picks whichever question tells it the most about **this** claim
- Asks about pre-authorisation first when the procedure needs it — never when it doesn't
- Updates the chance of rejection after every answer
- Stops once the verdict is settled: never under 3 questions, never over 8

3.6

questions on average

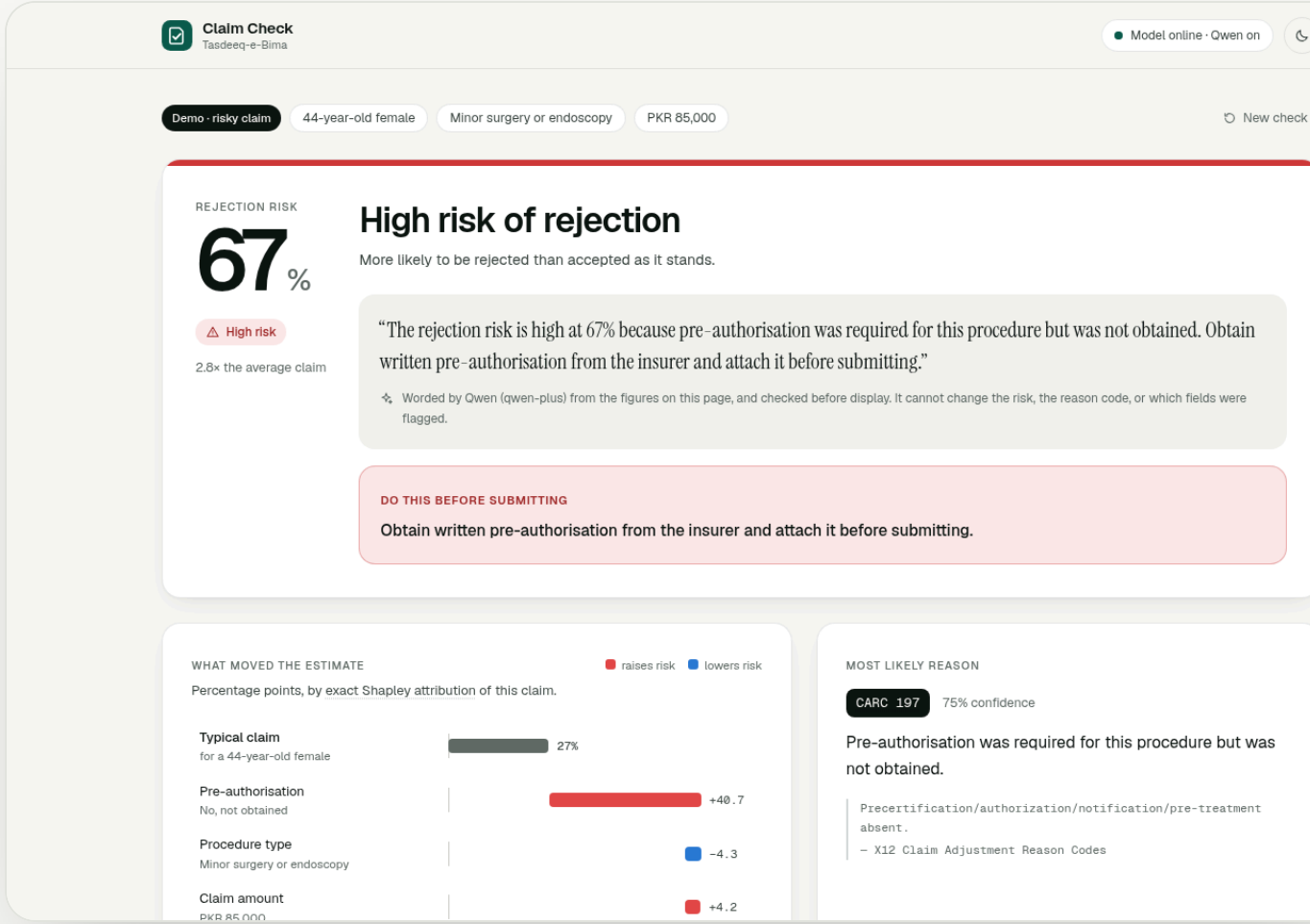
55

different question orders across 362 test claims

WHAT THE CLINIC SEES

A verdict it can act on

- Chance of rejection, with a **low / medium / high** verdict
- **Which field** is causing it, in points
- **Six insurer rules**: pre-auth, deadline, tariff, coverage, diagnosis fit, patient fit
- The official **rejection code** (X12 CARC) and the one thing to fix
- A plain-English summary



RESPONSIBLE AI

The model decides. Qwen only explains.

MACHINE-LEARNING MODEL

Computes the chance of rejection, the verdict and the likely reason

INSURER RULE TABLES

Check pre-auth, deadlines, tariffs and coverage — no AI involved

QWEN · ALIBABA CLOUD

Writes one plain sentence from the finished result

- Any reply quoting a number it wasn't given is **discarded** and a fixed template is shown
- No network? The template still works — the demo never breaks

RESULTS

Tested on claims it never saw

89%

of claims it cleared were accepted

72%

of rejections flagged before submission

56%

of “high risk” claims were rejected (average 24%)

0.85

AUC with every answer known (0.76 mid-check)

362 held-out simulated claims, used once. Verdict thresholds were fitted on a separate split, not chosen by hand.

HONEST LIMITS

What it does not do

- **Not trained on real claims** — no public dataset exists. Records are simulated (Synthea); the rules and rejection codes are real
- **Doesn’t replace staff** — it is a second check before submitting
- **Doesn’t submit claims** or talk to insurers
- **Doesn’t learn by itself** — staff corrections are logged, not auto-applied
- **No login or encryption yet** — built for a demo with synthetic data

ROADMAP

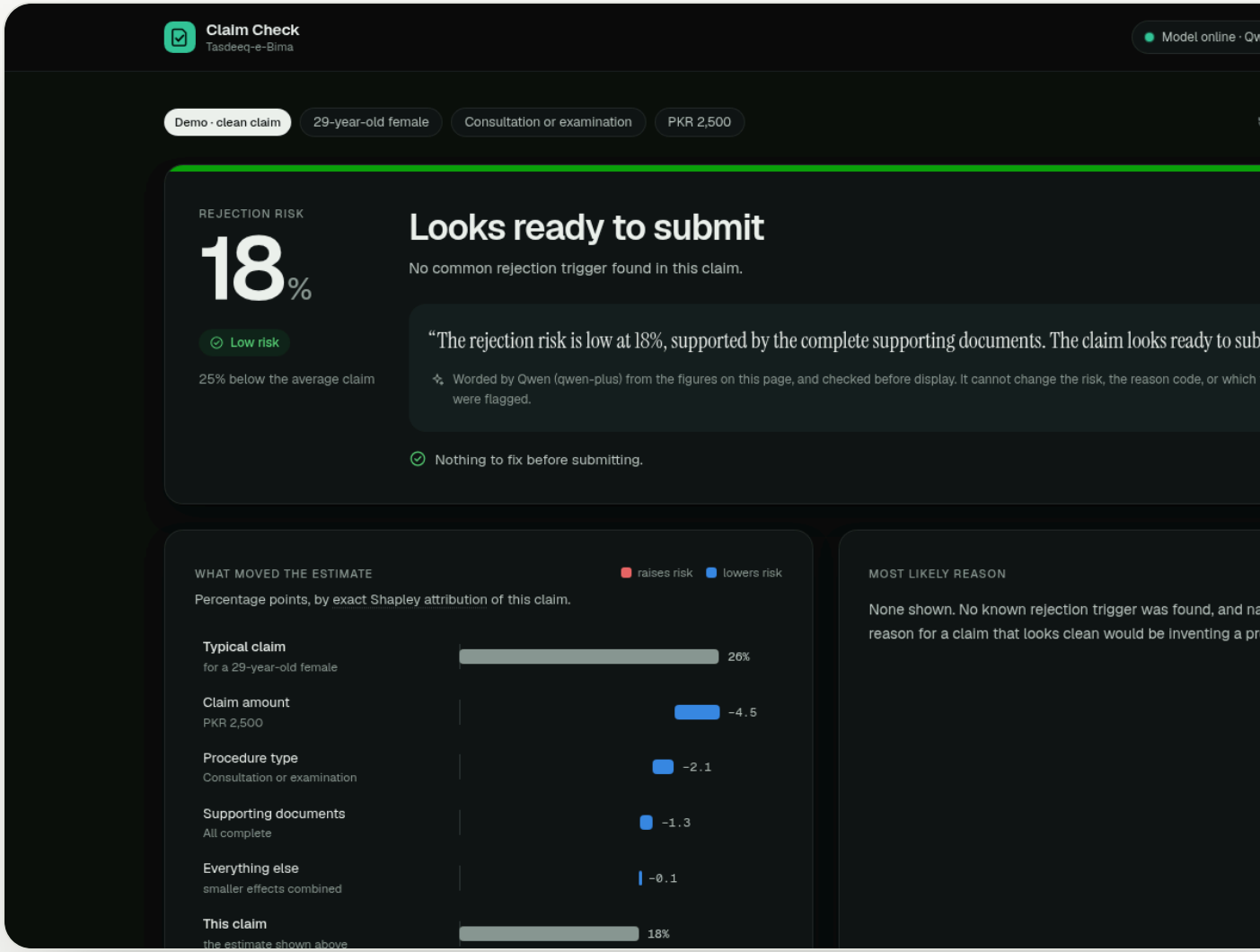
What we'd build next

- **Pilot** with a partner clinic or TPA — real claims, real outcomes
- **Real rejection reasons** read from insurers' remittance files
- **Per-insurer** rules and models
- **Urdu** and regional-language interface
- **Connect** to clinic billing software and national e-claims portals
- **Security** for real data: login, encryption, data agreements

CLAIM CHECK · TASDEEQ-E-BIMA

Catch the rejection before you submit.

We're looking for a **partner clinic or TPA** to pilot with real claims.



Mirza Ahsan Baig · Safee Akmal

Thank you

SOURCES

Where the figures come from

1. Premier Inc., “Claims Adjudication Costs Providers $25.7 Billion”, Feb 2025 (2023 data)
   premierinc.com/newsroom/blog/claims-adjudication-costs-providers-25-7-billion
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3. KFF, “Claims Denials and Appeals in ACA Marketplace Plans in 2023”
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4. Outlook Business, on the IRDAI Annual Report 2023-24, 30 Dec 2024
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   businessdailyafrica.com/bd/corporate/health/rural-private-hospitals-reject-sha-cover-over-sh22bn-debt-4934984
7. WHO & World Bank, Tracking Universal Health Coverage: 2025 Global Monitoring Report
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8. MarkNtel Advisors, Global Healthcare Denial Management Market 2025-2030 (market estimate)
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9. American Medical Association, 2024 Prior Authorization Physician Survey
   ama-assn.org/press-center/ama-press-releases/ama-survey-indicates-prior-authorization-wreaks-havoc-patient-care

Our own results: models/metadata.json and docs/model-card.md in the project repository (3,612 simulated claims, test split of 362).